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Electrical power regulating apparatus

Abstract

The invention relates to an apparatus for regulating electrical power in an electricity transmission network, the apparatus including: a DC contactor; a transmission network connector including live terminal(s) connected to live connection(s) and a neutral terminal connected to a neutral or earth of the electricity transmission network; switches connected to the DC contactor; electronic controlling devices coupled to the switches and control the switches to independently regulate electrical power on each of the live connection(s) and the neutral connection, the electronic controlling devices receive voltage reading of the live connection(s); calculate average of the voltage reading for the live connection(s); if the average is larger than an upper value, control the switches to reduce voltage on the live connection; if the average is less than a lower value, control the switches to increase voltage supplied on the live connection(s).

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Background/Summary

BACKGROUND OF THE INVENTION
(1) The present invention relates to an electrical power regulating apparatus, and in one particular example, an electrical power regulating apparatus for regulating power in an electrical transmission or distribution network.
DESCRIPTION OF THE PRIOR ART
(2) In a power transmission or distribution system, a power regulating device is implemented to regulate the supply of power over a wide range of load conditions. Currently, a static synchronous compensator (STATCOM) is typically used as a regulating device in alternating current (AC) transmission networks. A STATCOM for a distribution network are often referred to as dSTATCOM which uses electronic switching of energy storage banks to stabilise network supplied voltages.
(3) More recently, dSTATCOM devices are used to regulate power in three-phase electrical distribution networks. However, current three-phase dSTATCOM devices can only operate in a balanced-phased mode, in which all three phases are regulated together. This limitation restricts the efficacy, performance and functionality of the dSTATCOM.
(4) The reference in this specification to any prior publication (or information derived from it), or to any matter which is known, is not, and should not be taken as an acknowledgment or admission

or any form of suggestion that the prior publication (or information derived from it) or known matter forms part of the common general knowledge in the field of endeavour to which this specification relates.

SUMMARY OF THE PRESENT INVENTION

(5) In one broad form an aspect of the present invention seeks to provide an electrical power regulating apparatus configured to regulate electrical power in an electricity transmission network, the apparatus including: a DC contactor having DC terminals configured to be connected to a DC device; a transmission network connector configured to be connected to the electricity transmission network and including: at least one live terminal configured to be connected to the electricity transmission network; and a neutral terminal configured to be connected to at least one of a neutral or earth of the electricity transmission network; a plurality of switches connected to the DC contactor; at least one live connection connected to the switches and a respective live terminal; a neutral connection connected to the switches and the neutral terminal; a one or more filters configured to suppress noise on the at least one connection; and one or more electronic controlling devices coupled to the switches and being configured to control the switches to selectively connect the DC terminals and the at least one live connection and the neutral connection to thereby independently regulate electrical power on each of the at least one live connection and the neutral connection.

(6) In one embodiment the one or more electronics controlling devices are configured to control the switches to operate in a current mode or a voltage mode.

(7) In one embodiment the one or more electronics controlling devices are configured to control the switches to change between operating in a current mode and a voltage mode.

(8) In one embodiment, when in the current mode, the one or more electronics controlling devices are configured to control the switches so that each of the least one of the live terminals or the DC terminals have a predetermined current waveform.

(9) In one embodiment, when in the voltage mode, the one or more electronics controlling devices are configured to control the switches so that each of the least one live terminals or the DC terminals have a predetermined voltage waveform.

(10) In one embodiment, wherein the apparatus further includes a snubber circuit connected between the one or more filters and the DC contactor for suppressing noise.

(11) In one embodiment the neutral connection is decoupled from an earth.

(12) In one embodiment the switches include a plurality of silicon carbide MOSFET switches.

(13) In one embodiment the switches include a plurality of symmetrical half-bridge topology arms connecting to the at least one live connection and the neutral connection.

(14) In one embodiment the one or more filters include at least one of: an electromagnetic interference (EMI) choke; a differential-mode choke connected to the at least one live connection and the neutral connection for filtering noise; and a common-mode choke connected to the switches for suppressing interference.

(15) In one embodiment the switches have a plurality of capacitors configured to at least one of: smooth fluctuations in DC power; and store enough energy to complete each switching cycle with a full-phase offset range.

(16) In one embodiment the plurality of capacitors are connected to a failsafe mechanism to discharge the capacitors in the event of a fault.

(17) In one embodiment the capacitors are connected through an interleaving structure with a multi-layer printed circuit board (PCB).

(18) In one embodiment the interleaving structure includes one or more through-holes connecting a positive contact of one of the capacitors to one or more positive layers of the PCB and a negative contact of one of the capacitor to one or more negative layers of the PCB.

(19) In one embodiment the multi-layer PCB defines alternating polarity layers to cancel generated magnetic fields.

(20) In one embodiment the multi-layer PCB includes eight electrically conductive layers.
(21) In one embodiment the multi-layer PCB has at least two outer electrically conductive layers being negative layers.
(22) In one embodiment the failsafe mechanism includes at least one of: a hardware failsafe mechanism being configured to disconnect the DC contactor or the transmission network connector in the event of a fault; and a software failsafe mechanism, in the event of a fault, being configured to at least one of: rapidly connect and disconnect the capacitors to earth; and turn off the DC contactor or the transmission network connector.
(23) In one embodiment the one or more electronic controlling devices include a master controller and a slave controller.
(24) In one embodiment the DC device includes at least one of: a battery; a solar power generator; a hydrokinetic power generator; and a wind power generator.
(25) In one embodiment the transmission network connector includes at least one of: an AC contactor; an AC relay; and an AC circuit breaker for each of the at least one live terminal.
(26) In one embodiment the apparatus further includes a load connector configured to be connected to a load and including: at least one load terminal configured to be connected to the load; and a load neutral terminal configured to be connected to at least one of a neutral or earth of the load.
(27) In one embodiment the load connector includes a DC circuit breaker for each of the at least one load terminal.
(28) In one embodiment the apparatus further includes a communication interface connected to the one or more electronic controlling devices for communicating with an external device.
(29) It will be appreciated that the broad forms of the invention and their respective features can be used in conjunction and/or independently, and reference to separate broad forms is not intended to be limiting. Furthermore, it will be appreciated that features of the method can be performed using the system or apparatus and that features of the system or apparatus can be implemented using the method.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various examples and embodiments of the present invention will now be described with reference to the accompanying drawings, in which:—
(2) FIG. **1** is a schematic diagram of an example of an electrical power regulating apparatus;
(3) FIG. **2** is a flow chart of an example of an operation of an electrical power regulating apparatus;
(4) FIG. **3** is a schematic diagram of an example of a processing system;
(5) FIG. **4** is a schematic diagram of an example of an electrical power regulating apparatus;
(6) FIGS. **5**A and **5**B are schematic diagrams of capacitors on a printed circuit board of an electrical power regulating apparatus; and
(7) FIG. **6** is a schematic diagram of an example of switches of an electrical power regulating apparatus.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(8) An example of an electrical power regulating apparatus will now be described with reference to FIG. **1**.
(9) An electrical power regulating apparatus **100** configured to regulate electrical power in an electricity transmission network. The electrical power regulating apparatus **100** includes a DC contactor **110**, a transmission network connector **120**, a plurality of switches **130**, a one or more filters **140**, one or more electronic controlling devices **150**, at least one live connection **160**A, **160**B, **160**C, and a neutral connection **160**N.
(10) The DC contactor **110** having DC terminals **110**A, **110**B is configured to connect to a DC

device (not shown). The DC device may be DC power supply, such as at least one of a battery, a solar power generator; a hydrokinetic power generator, a wind power generator, or the like.

(11) The transmission network connector **120** is configured to be connected to the electricity transmission network, in this example, via a transmission bus **170**. The transmission network connector **120** includes at least one live terminal **120**A, **120**B, **120**C, and the live terminals **120**A, **120**B, **120**C are configured to be connected to the electricity transmission network. The transmission network connector **120** further includes a neutral terminal **120**N being configured to be connected to at least one of a neutral or earth of the electricity transmission network.

(12) One or more filters **140** are provided that are configured to suppress noise on the at least one connection **160**A, **160**B, **160**C. In this example, the transmission network connector **120** is coupled to the one or more filters **140** with at least one live connection **160**A, **160**B, **160**C and a neutral connection **160**N. The live connections **160**A, **160**B, **160**C are connected their respective live terminal **120**A, **120**B, **120**C, and the neutral connection **160**N is connected to the neutral terminal **120**N. However, it will be appreciated that other suitable configuration could be used.

(13) The plurality of switches **130** is connected to the DC contactor and provide onward connectivity to the transmission network connector, in this example by being connected to the one or more filters **140**, and specifically with the least one live connections **160**A, **160**B, **160**C and a neutral connection **160**N.

(14) The switches **130** are further coupled with the one or more electronic controlling devices **150**. The one or more electronic controlling devices **150** are configured to control the switches **130** to selectively connect the DC terminals **110**A, **110**B and the at least one live connection **160**A, **160**B, **160**C and the neutral connection **160**N to thereby independently regulate electrical power on each of the at least one live connection **160**A, **160**B, **160**C and the neutral connection **160**N.

(15) Accordingly, the one or more controlling devices **150** may be formed from any suitable controlling device that is capable of controlling the switches **130**, and could include a microprocessor, microchip processor, logic gate configuration, firmware optionally associated with implementing logic such as an FPGA (Field Programmable Gate Array), or any other electronic device, system or arrangement. Furthermore, for ease of illustration the remaining description will refer to an electronic controlling device, but it will be appreciated that multiple controlling devices could be used, with controlling distributed between the devices as needed, and that reference to the singular encompasses the plural arrangement and vice versa.

(16) An example of operation of the electrical power regulating apparatus **100** will now be described with reference to FIG. **2**.

(17) In this example, the electrical power regulating apparatus **100** connects to the electricity transmission network. At step **200**, the one or more live terminals **120**A, **120**B, **120**C of the electrical power regulating apparatus **100** receives electrical power from the electricity transmission network, and the neutral terminal **120**N is connected to the neutral or earth of the electricity transmission network. In one example, each of the one or more live terminals **120**A, **120**B, **120**C receives a respective phase of the electrical power from the electricity transmission network.

(18) The live connections **160**A, **160**B, **160**C are connected to the respective live terminals **120**A, **120**B, **120**C to carry a respective phase of the AC power. At step **210**, the AC power carried on the live connections **160**A, **160**B, **160** and the neutral connection **160**N are passed through to the one or more filters **140** for filtering noise and/or interference of the AC power.

(19) The filtered AC power is passed to the switches **130** with the one or more live connection **160**A, **160**B, **160**C and the neutral connection **160**N. At step **220**, the electronic controlling device **150** controls the switches **130** of the one or more live connection **160**A, **160**B, **160**C and the neutral connection **160**N. The switches **130** are controlled to selectively connect the DC terminals **110**A, **110**B and the at least one live connection **160**A, **160**B, **160**C and the neutral connection **160**N, so that electrical power on each of the one or more live connection **160**A, **160**B, **160**C and the neutral

connection **160**N are independently regulated. This allows DC power to be output to the DC terminals **110**A, **110**B at step **230**, for example to charge a battery.
(20) Alternatively, this allows DC power supplied to the DC terminals from the DC device to be used to modulate power on the live connection **160**A, **160**B, **160**C and the neutral connection **160**N at step **240**. For example, the electrical power regulating apparatus **100** can independently regulates the electrical power characteristics on each of the one or more live connections **160**A, **160**B, **160**C and the neutral connection **160**N, for example by adjusting a phase angle, magnitude or condition of the voltages or current on the connections, which can in turn be used to accommodate unbalanced voltage or current input on each of the live connections and allows adjustment of the voltage and/or current on each of the live connections thereby conditions and/or regulates the power. As the electrical power regulating apparatus **100** is able to independently regulate the electrical power on each of the one or more live connections **160**A, **160**B, **160**C and the neutral connection **160**N, this allows the electrical power regulating apparatus **100** to operate with an unbalanced load or as an unbalanced supply. This also allows the electrical power regulating apparatus **100** to be implemented in any one of a three-phase, two-phase and a single-phase system, and operates to regulate voltage and/or current of the system.
(21) It should be appreciated that the electrical power regulating apparatus **100** herein described regulates for AC-to-DC power system is for exemplary only. The electrical regulating apparatus **100** is capable of regulating DC-to-AC power systems, AC-to-AC power systems or DC-to-DC power systems.
(22) A number of further features will now be described.
(23) The electronics controlling device is configured to control the switches to operate in a current mode or a voltage mode. Additionally, the electronics controlling device is configured to control the switches to change between operating in a current mode and a voltage mode. This allows the electrical power regulating apparats to provide a suitable voltage or current to meet an output requirement and transitions between the modes when the requirement changes.
(24) Accordingly, when in the current mode, the electronics controlling device is configured to control the switches so that each of the least one live terminals or the DC terminals have a predetermined current waveform. Similarly, when in the voltage mode, the electronics controlling device is configured to control the switches so that each of the least one live terminals or the DC terminals have a predetermined voltage waveform. This allows the electrical power regulating apparats to provide voltage or current waveforms that meets the requirements of the DC device or the electricity transmission network.
(25) The electrical power regulating apparatus may further include a snubber circuit connected between the one or more filters and the DC contactor for suppressing noise.
(26) In one example, the neutral connection is decoupled from an earth, such as an earth of the electricity transmission network, which can be achieve using one or more capacitors, or the like. This allows the neutral connection to be regulated or modulated, which in turn provides greater flexibility in terms of the overall control the system can provide.
(27) The switches may be a plurality of symmetrical half-bridge topology arms that connect to the at least one live connection and the neutral connection, which allows selectable switching of the live connections and the neutral connection. Additionally, the switches may be silicon carbide MOSFET switches, which is advantageously allows the switches to be physically compact.
(28) In one example, the one or more filters may be at least one of: an electromagnetic interference (EMI) choke; a differential-mode choke connected to the at least one live connection and the neutral connection for filtering noise; and a common-mode choke connected to the switches for suppressing interference. The one or more filters advantageously eliminate the noise or interference from the electricity transmission network or the switches.
(29) The switches may have a plurality of capacitors configured to store ample energy to complete each switching cycle with a full-phase offset range and hence smooth fluctuations in DC power, so

that it facilitates in protecting the DC devices by providing DC power with minimal fluctuations. The capacitors may be any one of film capacitors, ceramic capacitors and electrolytic capacitors. It should be appreciated that other types of capacitors can also suitable.
(30) The plurality of capacitors may be connected through an interleaving structure with a multi-layer PCB to further minimise noise caused by the PCB tracks. In one example, the interleaving structure includes one or more through-holes connecting a positive contact of one of the capacitors to one or more positive layers of the PCB and a negative contact of one of the capacitor to one or more negative layers of the PCB. Additionally, the multi-layer PCB may define alternating polarity layers so as to cancel magnetic fields generated by each layer of PCB. In one example, the multi-layer PCB including eight electrically conductive layers having electrically conductive negative layers, wherein two of the negative layers are outer layers and alternating polarities in between layers.
(31) In one example, the plurality of capacitors may be further connected to a failsafe mechanism to discharge the capacitors in the event of a fault. The fault can be over-current, voltage overshoot, loss of power, controlling device failure and/or communication failure.
(32) The failsafe mechanism may be a hardware failsafe mechanism and/or a software failsafe mechanism. In this example, the hardware failsafe mechanism is configured to disconnect the DC contactor or the transmission network connector in the event of the fault, whereas the software failsafe mechanism is configured to rapidly connect and disconnect the capacitors to earth and/or turn off the DC contactor or the transmission network connector. The failsafe mechanism provides protection to the electronics of the apparatus, such as the MOSFET switches, microcontrollers, from being damaged by over-current or voltage overshoot.
(33) The electronic controlling device may include a master controller and a slave controller.
(34) The transmission network connector may include at least one of an AC contactor; an AC relay; and an AC circuit breaker for each of the at least one live terminal.
(35) Additionally the electrical power regulating apparatus may include a load connector for connecting to a load. The load connector may include at least one load terminal connected to the load and a load neutral terminal connected to at least one of a neutral or earth of the load. Similarly, the load connector may include a DC circuit breaker for each of the at least one load terminal.
(36) The electrical power regulating apparatus may further include a communication interface connected to the electronic controlling device for communicating with an external device. The communication interface may be WAN, Bluetooth, WLAN and/or suitable serial ports to allow the electrical power regulating apparatus to be controlled or configured by an external computer.
(37) An example of an electronic controlling device will now be described with reference to FIG. **3**.
(38) In this example, the electronic controlling device **160** includes at least one microprocessor **300**, a memory **301**, an optional input/output device **302**, such as a keyboard and/or display, an interface **303**, interconnected via a bus **304** as shown. In this example the interface **303** can be utilised for connecting the electronic controlling device **160** to peripheral devices, such as communications networks, or the like.
(39) In use, the microprocessor **300** executes instructions in the form of applications software stored in the memory **301** to allow the required processes to be performed, including controlling the electronic controlling device **160**. The applications software may include one or more software modules, and may be executed in a suitable execution environment, such as an operating system environment, or the like.
(40) Accordingly, it will be appreciated that the electronic controlling device **160** may be formed from any suitable control system and could include be any electronic processing device such as a microprocessor, microchip processor, logic gate configuration, firmware optionally associated with implementing logic such as an FPGA (Field Programmable Gate Array), or any other electronic device, system or arrangement.
(41) However, it will be appreciated that the above described configuration assumed for the

purpose of the following examples is not essential, and numerous other configurations may be used. It will also be appreciated that the partitioning of functionality between the different processing systems may vary, depending on the particular implementation.
(42) An example of the electrical power regulating apparatus will now be described in more detail with reference to FIG. **4**.
(43) An electrical power regulating apparatus **400** configured to regulate electrical power in an electricity transmission network. The electrical power regulating apparatus **400** includes one or more DC contactors **410**, a plurality of capacitors **431**, a plurality of switches **430**, a common-mode (CM) choke **441**, one or more differential-mode (DM) chokes **442**, an EMI choke **443**, a grid connector, and a load connector. The grid connector includes an AC connector **421**, an AC relay **422**, AC circuit breakers **423**, and three AC terminals **420**A, **420**B, **420**C and an AC neutral connector **420**N. The load connector includes load circuit breakers **471** and three load terminals **470**A, **470**B, **470**C and the load neutral connector **470**N. The electrical power regulating apparatus **400** also includes three connections **460**A, **460**B, **460**C, and a neutral connection **460**N. The three connections **460**A, **460**B, **460**C are connected to the respective live connection with the AC terminals **420**A, **420**B, **420**C. The neutral connection is connected to the respective grid/live side neutral connection. Connections made post an electrical power regulating apparatus are connected in accordance with local rules, regulations and standards.
(44) The DC contactor **410** having DC terminals **410**A, **410**B. The DC terminals **410**A, **410**B are configured to connect to a battery and a low voltage (LV) solar power generator. In this example, the DC contactor **410** includes at least one mechanical switch for switching both terminals **410**A, **410**B. Mechanical switches are robust and reliable, and can be configured to have a high breaking capabilities in voltage and current to improve electrical safety.
(45) In one embodiment, a safety interlocking mechanism is implemented. The safety interlocking mechanism may include interlocking multiple AC contactors and/or interlocking both the AC and DC contactors. The safety interlocking mechanism may utilizes an on-board processor, such as one of the controlling devices **451**, **452**, and a supplementary external processor to switch on the supply. The on-board processor monitors the status of the electrical power regulating apparatus **400** while the external processor monitoring at safety considerations, such as battery voltage and/or battery polarity. Both processors crosscheck each other before operating the contactor.
(46) A DC bus sensor **410**C is implemented to monitor the current at the DC terminals **410**A, **410**B. The DC bus sensor **410**C is coupled to the controlling device **451** and allows leakage current or residual current to be detected. In an example, a Sigma-Delta configuration is used for the DC bus sensor **410**C, which provides high voltage isolation and accuracy in sensing.
(47) DC Capacitors
(48) The DC contactor **410** is connected to the switches **430** via the capacitors **431**. In this example, the capacitors **431** are polypropylene capacitors. The polypropylene capacitors are capable of operating with relatively high ripple currents, which allows the users to control higher order harmonics without decreasing capacitor life. Polypropylene capacitor also have self-healing capabilities, which allows the size of the capacitors to be reduced for providing a required capacitance and accommodating transient voltage.
(49) In this example, there are twelve capacitors in the electrical power regulating apparatus **400**. The capacitors are mounted on a multi-layer printed circuit board (PCB) in a three-by-four array arrangement. The tracks connecting the capacitors are on the multi-layer PCB with an interleaving structure to minimise parasitic inductance. The interleaving structure is described in more detail with reference to FIGS. **5**A and **5**B. As illustrated in FIG. **5**A, the twelve capacitors are placed on the board **520** in an alternating polarity manner. Accordingly, a positive contact of a capacitor **511** is positioned adjacent to a negative contact of a capacitor **512**. Furthermore, the board **520** has eight layers and each layer is plated with copper which makes the layer conductible, thereby minimising the ‘trace’ inductance effect. The layers are connected to either the positive or the negative of the

capacitor contacts, which makes an alternating polarity pattern of the layers.
(50) Referring to FIG. **5**B, the capacitor **510** has a positive contact **510***a* and a negative contact **510***b*. The positive contact **510***b* is in contact with the second layer **522**, the fifth layer **525** and the seventh layer **527** by a through-hole (also called 'via') structure **520***a* of the board **520**. The negative contact **510***b* is soldered on the first layer **521**, whilst in contact with other layers **523**, **524**, **526** and **528** with a through-hole structure **520***b*. Similarly, capacitors **511**, **512** are connected to the board **520** with through-hole structures **520***c*, **520***d* and **520***e*, **520***f* as illustrated in FIG. **5**B. In one example, the interleaved structure uses eight layers with approximately 2 oz copper per layer, and the layers are separated with FR4 PCB material with a thickness of approximately 0.4 mm. It should be appreciated that other suitable materials and/or thickness may be used depending on different system requirement.
(51) The interleaving structure allows the parasitic inductance to be reduced as the current is distributed across multiple layers. There are also no 'tracks' on the layers which induces inductance. The parallel layers with alternating polarity may increase the parasitic capacitance to complement the capacitors. As described, there is an alternating plate pattern between positive and negative, thereby minimising parasitic inductance and increasing total capacitance. This also improves the equivalent series resistance (ESR) and temperatures as there is less eddy current circulation (phenomenon where induced currents from magnetic fields interact with traces/wires). With a more stable temperature the change in capacitance due to temperature during operation is reduced.
(52) In this example, the top layer **521** and bottom layer **528** are negative, so that the board **520** has a minimal electromagnetic interference (EMI) and therefore an improvement in the electromagnetic compatibility (EMC). The interleaving structure also reduces EMI currents that are induced within the board **520**, and thereby reducing overall magnitude of noise within the board **520**.
(53) The electrical power regulating apparatus **400** also includes a failsafe system **433** coupling the capacitors **432** to earth and controls the discharging of the capacitors in the event of a fault. The failsafe system **433** includes a software failsafe and a hardware failsafe mechanism. When a voltage abnormality is detected, the software failsafe mechanism switches the capacitors on and off rapidly to safely discharge the capacitors. Moreover, if power is lost, the hardware failsafe mechanism opens the DC and AC contactors. The hardware failsafe mechanism also includes an external DC battery fuse and an external AC grid fuse.
(54) Switches
(55) The capacitors **431** are connected to the switches **430** for modulating the each of the three connections **460**A, **460**B, **460**C, and the neutral connection **460**N. FIG. **6** shows an example of the switches **430**, which includes four symmetrical half bridge topology arms **430**A, **430**B, **430**C, **430**N connecting to each of the three connections **460**A, **460**B, **460**C and the neutral connection **460**N. Each symmetrical half bridge topology arms includes two MOSFET switches. In this example, the MOSFET switches are silicon carbide switches, particularly, 1200V 55 A SCT3040 MOSFET in compliant with the ROHM standard.
(56) The switching characteristics, such as rise time, fall time, and dead-time, of the switches **430** can be configured by hardware design and/or the controlling devices **451**, **452** to optimise the performance. The switches **430** can operate in higher switching frequencies (approximately 100 kHz) due to a reduction of the primary side to secondary side capacitance as the specific isolation devices are used. This also allows high voltage isolation barrier to increase noise immunity and provide high voltage protection between primary side and secondary side. The switches **430** may further include a snubber circuit **433** to reduce the voltage changes over time (dv/dt) to meet the EMI/EMC requirements.
(57) CM Choke
(58) The CM choke **441** is coupled to the switches **430** to suppress and/or reject high frequency common mode current. The live connections **460**A, **460**B, **460**C and the neutral connection **460**N

are connected to the CM choke **441** and passed through a core. The core of the CM choke **441** may be a powder core and can be configured to operate in high frequency with low loss. The CM choke **441** is also configured to operate in wider temperature range condition while maintaining low energy losses. Furthermore, in one example, the CM choke **441** may be insulated from an outer chassis of the electrical power regulating apparatus **400**, so that high frequency magnetic coupling to the surrounding equipment is minimised. The windings of the CM choke **441** may be of a non-interleaved structure, thereby reducing coupling noise propagating to adjacent conductors. Additionally, the winding may be copper flat bar winding, which minimises copper losses in low and high frequencies. Flat bar winding may also reduce skin effect when compared to round windings. The increased surface area of the flat bar winding may help with heat dissipation of the CM choke **441**.

(59) DM Choke

(60) The CM choke **441** is coupled to a DM choke **442**, which acts like a bandpass filter and smooths high frequency from the PWM generated by the switches **430**. The DM choke **442** may include individual differential mode inductors that are built of EE cores. The EE cores are compact and easier to manufacture when compared to toroidal cores. The EE core may also shields from high frequency noise and stray magnetic fields. Additionally, the core of the DM choke **442** may be made of a high saturation-point powder core material, so that the DM choke **442** can be configured to operate in high frequency with low loss. This allows the DM choke to have a consistent filter performance, and the inductor to have an overrated capacity a short term. The DM choke **441** can be configured to operate in wider temperature range condition while maintaining low energy losses. Furthermore, the DM choke **442** may be insulated from an outer chassis of the electrical power regulating apparatus **400**, so that high frequency magnetic coupling to the surrounding equipment is minimised.

(61) The winding of the DM choke **442** includes copper film winding, which may be thin and flat copper film, to minimise copper loss in low and high frequencies. Copper film winding may also reduce skin effect when compared to round windings. The increased surface area of the film winding may help with heat dissipation of the DM choke **442**. The use of film winding also allows an increase in the number of turns without compromising in size and/or skin effect losses.

(62) EMI Filter

(63) The DM choke **442** is coupled to an EMI filter **443**, which filters out high frequency electromagnetic interference. This may help preventing any stray common mode current from exporting to the grid, any common mode current from grid impacting the control circuitry. The EMI filter **443** includes a ring core with nano-crystalline material. Solid enamelled copper wire may be used around core material to minimise copper losses, particularly at 50 Hz. The EMI filter **443** includes symmetrical windings to minimise leakage inductance. The EMI filter **443** may also include Y capacitors (Y2 rated) in its PCB design.

(64) It should be appreciated that current and/or voltage measurements may be implemented at any point in the electrical power regulating apparatus **400**. In one example, the AC current and voltage measurements are implemented between the DM choke **442** and the EMI choke **443**. The AC current may be measured with a LEM (100−p) 100 A high precision hall-effect sensor.

(65) The electrical power regulating apparatus **400** includes Class X capacitors (X2 rated) between the DM choke **442** and the EMI choke **443**. The X capacitors can be useful for high voltage applications to minimise high voltage transients. This also allows shielding signal traces from power traces on the PCB and also provide protection from primary side to secondary side. As illustrated in FIG. **4**, an earth link is provided at this point for the neutral connection **460**N through a decoupling capacitor.

(66) Load Contactor

(67) In this example, the electrical power regulating apparatus **400** includes the load connector, so that the electrical power regulating apparatus **400** can provide UPS functionality while connected

to the grid. The load connector has a **63**A circuit breaker **471** for each live connection **460**A, **460**B, **460**C. The connectors may be Amphenol connectors. As the electrical power regulating apparatus **400** is able to compensate the harmonic components to match the load component, this allows the harmonic content to be managed and not passed through to the grid. This also provides the load with a "clean" AC source. The electrical power regulating apparatus **400** can operate with an unbalanced load condition as the neutral connection **460**N can be modulated.

(68) This allows the electrical power regulating apparatus **400** to be a source to the load regardless to whether the load is a three-phase, two-phase or a single-phase load. This is particularly useful for off grid loads with large induction machines, such as farming equipment, conveyor belts, fridges etc.

(69) AC Contactor

(70) The AC contactor **421** is coupled to the EMI filter **443** for connecting to the grid. The AC contactor **421** may be a Semmens three-phase (AC-3 rated) contactor with 38 kW breaking capacity which has low power consumption, such as less than 4 W. The AC contactor **421** is also robust, reliability and able to break high fault currents while providing short switch on and off response times. The AC contactor **421** may include internal sensors in communications with at least one of the controlling devices **451**, **452**, so that the AC contactor operational status is monitored and controlled.

(71) AC Relay

(72) An AC relay **422**, **422***a*, **422***b* are coupled between the AC contact **421** and the grid to provide an additional isolation. As illustrated in FIG. **4**, the neutral connection **460**N are separately coupled to two AC relay **422***a*, **422***b*. The AC relay **422**, **422***a*, **422***b* are in communication with at least one of the controlling devices **451**, **452**. Additionally, the AC contactor **421** and the AC relay **422** are controlled by different controlling devices **451**, **452**. This allows the AC contactor **421** and the AC relay **422** to perform interlocking which provide additional safety. Similarly, the two AC relay **422***a*, **422***b* are also controlled by different controlling devices **451**, **452** to provide additional safety to the neutral connection **460**N.

(73) Circuit Breaker

(74) Similar to the load connection circuit breakers **471**, circuit breakers **423** are coupled to each of the live connections to form the isolation barrier between the AC terminals **420**A, **420**B, **420**C and the AC contactor **421**. This provides additional overcurrent protection to the electrical power regulating apparatus **400**. The circuit breakers **423** are rated at **63**A to allow operations in various temperatures and no prematurely breakage due to temperature.

(75) MCU

(76) The controlling devices **451**, **452** may be two on-board microprocessors. A first microprocessor **451** includes executable programs to control the operation of the apparatus **400**, and a second microprocessor **452** is functioned as a safety interlocking controller. An analogue measurement **451***a* and a temperature measurement **452***a* may be coupled to the microprocessors **451**, **452** respectively. Furthermore, a communication interface **453** is also connected to the microprocessors **451**, **452**. The communication interface **453** may include one or more interface for Modbus (TCP/IP), USB, RS232/485, Ethernet, demand response mode (DRM) communications. This allows external processors to communicate with the apparatus **400** for monitoring, controlling and/or configuration the apparatus **400**.

(77) The electrical power regulating apparatus provides at least the following advantages: Fast response, often under 20 milliseconds Continuous control of voltage Addresses flicker issues Addresses voltage unbalance issues Low power losses Capable of generating a 415V three-phase output and interface a three-phase system to one battery. This provides cost advantages in that only one battery and battery management system per apparatus is needed. This also results in a smaller unit as the higher modulation frequency made possible by using MOSFETS. Smaller size means smaller cabinets, saving both space and materials. Being a four-quadrant device, enables it to be

flexible to manage both active and reactive power. This makes it advantageous in managing battery energy storage systems, solar photovoltaic (PV) installations and distribution system voltages Reliable as the capacitors and other components are all high-quality components as no electrolytic capacitors are used. The operation may be automatic based on the program in it, with little need for maintenance and servicing.
(78) Throughout this specification and claims which follow, unless the context requires otherwise, the word “comprise”, and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated integer or group of integers or steps but not the exclusion of any other integer or group of integers. As used herein and unless otherwise stated, the term “approximately” means±20%.
(79) It must be noted that, as used in the specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a support” includes a plurality of supports. In this specification and in the claims that follow, reference will be made to a number of terms that shall be defined to have the following meanings unless a contrary intention is apparent.
(80) It will of course be realised that whilst the above has been given by way of an illustrative example of this invention, all such and other modifications and variations hereto, as would be apparent to persons skilled in the art, are deemed to fall within the broad scope and ambit of this invention as is herein set forth.

Claims

1. An electrical power regulating apparatus configured to regulate electrical power in an electricity transmission network, including: a) a DC contactor having DC terminals configured to be connected to a DC device; b) a transmission network connector configured to be connected to the electricity transmission network and including: i) at least one live terminal configured to be connected to the electricity transmission network; and, ii) a neutral terminal configured to be connected to at least one of a neutral or earth of the electricity transmission network; c) a plurality of switches connected to the DC contactor; d) at least one live connection connected to the switches and a respective live terminal; e) a neutral connection connected to the switches and the neutral terminal; f) a one or more filters configured to suppress noise on the at least one connection; and, g) one or more electronic controlling devices coupled to the switches and being configured to control the switches to selectively connect the DC terminals and the at least one live connection and the neutral connection to thereby independently regulate electrical power on each of the at least one live connection and the neutral connection.
2. The electrical power regulating apparatus according to claim 1, wherein the one or more electronics controlling devices are configured to control the switches to operate in a current mode or a voltage mode.
3. The electrical power regulating apparatus according to claim 2, wherein, when in the current mode, the one or more electronics controlling devices are configured to control the switches so that each of the least one of the live terminals or the DC terminals have a predetermined current waveform.
4. The electrical power regulating apparatus according to claim 2, wherein, when in the voltage mode, the one or more electronics controlling devices are configured to control the switches so that each of the least one live terminals or the DC terminals have a predetermined voltage waveform.
5. The electrical power regulating apparatus according to claim 1, wherein the one or more electronics controlling devices are configured to control the switches to change between operating in a current mode and a voltage mode.
6. The electrical power regulating apparatus according to claim 1, wherein the apparatus further includes a snubber circuit connected between the one or more filters and the DC contactor for

suppressing noise.
7. The electrical power regulating apparatus according to claim 1, wherein the neutral connection is decoupled from an earth.
8. The electrical power regulating apparatus according to claim 1, wherein the switches include a plurality of silicon carbide MOSFET switches.
9. The electrical power regulating apparatus according to claim 1, wherein the switches include a plurality of symmetrical half-bridge topology arms connecting to the at least one live connection and the neutral connection.
10. The electrical power regulating apparatus according to claim 1, wherein the one or more filters include at least one of: a) an electromagnetic interference (EMI) choke; b) a differential-mode choke connected to the at least one live connection and the neutral connection for filtering noise; and, c) a common-mode choke connected to the switches for suppressing interference.
11. The electrical power regulating apparatus according to claim 1, wherein the switches have a plurality of capacitors configured to at least one of: a) smooth fluctuations in DC power; and, b) store enough energy to complete each switching cycle with a full-phase offset range.
12. The electrical power regulating apparatus according to claim 11, wherein the capacitors are connected through an interleaving structure with a multi-layer printed circuit board (PCB).
13. The electrical power regulating apparatus according to claim 12, wherein the interleaving structure includes one or more through-holes connecting a positive contact of one of the capacitors to one or more positive layers of the PCB and a negative contact of one of the capacitor to one or more negative layers of the PCB.
14. The electrical power regulating apparatus according to claim 12, wherein the multi-layer PCB at least one of: a) defines alternating polarity layers to cancel generated magnetic fields; b) includes eight electrically conductive layers; and c) has at least two outer electrically conductive negative layers.
15. The electrical power regulating apparatus according to claim 1, wherein the plurality of capacitors are connected to a failsafe mechanism to discharge the capacitors in the event of a fault.
16. The electrical power regulating apparatus according to claim 15, wherein the failsafe mechanism includes at least one of: a) a hardware failsafe mechanism being configured to disconnect the DC contactor or the transmission network connector in the event of a fault; and, b) a software failsafe mechanism, in the event of a fault, being configured to at least one of: i) rapidly connect and disconnect the capacitors to earth; and, ii) turn off the DC contactor or the transmission network connector.
17. The electrical power regulating apparatus according to claim 1, wherein at least one of: a) the one or more electronic controlling devices include a master controller and a slave controller; b) the DC device includes at least one of: i) a battery; ii) a solar power generator; iii) a hydrokinetic power generator; and, iv) a wind power generator; and, c) the transmission network connector includes at least one of: i) an AC contactor; ii) an AC relay; and, iii) an AC circuit breaker for each of the at least one live terminal.
18. The electrical power regulating apparatus according to claim 1, wherein the apparatus further includes a load connector configured to be connected to a load and including: i) at least one load terminal configured to be connected to the load; and, ii) a load neutral terminal configured to be connected to at least one of a neutral or earth of the load.
19. The electrical power regulating apparatus according to claim 18, wherein the load connector includes a DC circuit breaker for each of the at least one load terminal.
20. The electrical power regulating apparatus according to claim 1, wherein the apparatus further includes a communication interface connected to the one or more electronic controlling devices for communicating with an external device.